

Supplementary Material — OAAS Framework (Applied Acoustics submission)

This PDF provides compact, reader-friendly versions of Tables S1–S4. Full-resolution CSV tables are provided via the project repository (see Data availability statement).

Table S1. OAAS distances for vocalization anchors, functional music stimuli, and challenge reference stimuli (compact view).

Shown: top-10 most POS-biased (lowest $\Delta = d_{\text{POS}} - d_{\text{NEG}}$) and top-10 most NEG-biased (highest Δ). Distances are computed relative to vocalization-derived POS/NEG reference centroids in the joint OAAS embedding.

file	domain	OAAS1	OAAS2	OAAS3	d_pos_euclid	d_neg_euclid	delta_euclid
W7_07.wav	music	1.915285260665352	-2.674754383630324	-1.5536603353499985	5.435341786395151	3.1729349007755183	-2.2624068856196327
Negative_02_part3.wav	challenge_reference	-0.0030316027199865	-0.5184714444546679	-3.056790787262164	4.054280772841859	2.0019077431784424	-2.052373029663417
Negative_03.wav	challenge_reference	1.877633713339855	-0.5952803237737669	-3.4916068900313624	4.926142143957697	2.878213944578974	-2.047928199378723
W8_06.wav	music	0.4675865255883913	-1.2495465712761242	-0.5988446839180059	3.426564039525259	1.4105395751205871	-2.016024464404672
W6_05.wav	music	0.4675865255883913	-1.2495465712761242	-0.5988446839180059	3.426564039525259	1.4105395751205871	-2.016024464404672
W4_04.wav	music	-0.0225159753839707	-0.6016116350690116	-0.762103047825078	2.773296456663986	0.8226819629355578	-1.950614493728428
W2_04.wav	music	-0.0225159753839707	-0.6016116350690116	-0.762103047825078	2.773296456663986	0.8226819629355578	-1.950614493728428
W1_04.wav	music	-0.442746035370773	-1.0783481122644452	-0.7821415887360239	3.2303814074046984	1.3982810960882668	-1.832100311316432
W10_04.wav	music	-0.442746035370773	-1.0783481122644452	-0.7821415887360239	3.2303814074046984	1.3982810960882668	-1.832100311316432
W3_05.wav	music	-0.1635679048250887	0.2168373706419207	-1.5717503166116769	2.458061684679816	0.6561427455493342	-1.8019189391304813
ShortReunion__SW__02.wav	vocalization	0.3349752384611582	3.092357559691321	0.2174224385354086	1.22747163022476	3.352509201101768	2.125037570877008
ShortReunion__SW__01.wav	vocalization	0.3374940284454479	3.1051500427349463	0.2221975053199616	1.2403098077658483	3.3661510080385995	2.1258412002727507
MissedNursing__SW__02.wav	vocalization	-0.1071526877108738	2.728327665559757	0.1405980155977676	0.7196650141086373	3.0194758665479657	2.2998108524393284
MissedNursing__SW__01.wav	vocalization	-0.113320859891498	2.740464094052476	0.1468041776667622	0.7306889202822652	3.033832834419855	2.3031439141375896
BeforeNursing__SW__01.wav	vocalization	-0.2922050822597513	3.0221539943609432	0.1409542140994785	0.996665498578576	3.31312287790828	2.3164573793297043
BeforeNursing__SW__02.wav	vocalization	-0.2954382376960834	3.012921388784592	0.1374162268421566	0.9873727709769248	3.3040068225686614	2.3166340515917367
AfterNursing__SW__01.wav	vocalization	-0.082980563405053	2.855170533451856	0.276666356461388	0.867599028287396	3.188876748620112	2.3212777203327155
AfterNursing__SW__02.wav	vocalization	-0.0831842857693972	2.841607223418773	0.2734965236174368	0.8539305812188668	3.175484673213184	2.321554091994317
Huddling__SW__02.wav	vocalization	-0.2823655191520475	2.2828937199085586	0.1183489062674955	0.2581732865240815	2.652845883829655	2.3946725973055734
Huddling__SW__01.wav	vocalization	-0.278840852667629	2.266418880509072	0.1134171055007872	0.2410282923381518	2.6357258420232563	2.3946975496851044

Table S2. PCA explained variance ratios (OAAS1–OAAS3).

PC	Explained variance (%)	Cumulative variance (%)
PC1	45.31	45.31
PC2	28.72	74.03
PC3	9.98	84.01

Table S3. PCA loadings for the nine OAAS core features.

Variable	PC1	PC2	PC3
se_mean	0.1218	0.5356	0.0208
se_std	0.4628	0.0067	0.0415
se_p95	0.3976	0.3362	-0.072
flatness_mean	0.474	-0.0908	0.0993
flatness_std	0.4076	-0.2364	-0.2829
flatness_p95	0.4339	-0.0584	-0.1982
te_rms	0.1777	-0.2179	0.9069
mse_env	-0.0339	0.4677	0.1437
harm_ratio	-0.0335	-0.516	-0.1429

Table S4. Hypothesized SoundWel-to-QBA conceptual mapping layer.

This table is provided as an interpretive (hypothesis-driven) semantic layer and does not represent a validated affective outcome measure.

SoundWel category	QBA candidate labels	OAAS1	OAAS2	OAAS3	d_to_OAAS_POS	d_to_OAAS_NEG	delta_POS_minus_NEG	OAAS bias
AfterNursing	content, relaxed, calm	-0.083082424587225	2.848388878435314	0.2750814400394124	2.6548	3.1822	-0.5274	POS-biased
Barren	bored, uneasy, frustrated	-1.303691811927154	0.6592986752845327	0.0282566506705754	1.1842	2.1011	-0.9169	POS-biased
BeforeNursing	expectant, alert, interested	-0.2938216599779173	3.017537691572768	0.1391852204708175	2.8168	3.3086	-0.4918	POS-biased
Castration	distressed, fearful, agitated	0.6483016283046137	3.341971560180471	-0.0037186006708625	3.2582	3.5184	-0.2602	POS-biased
Crushing	distressed, fearful, agitated	0.4278019729382152	2.9072980574108103	0.0909665741682079	2.7805	3.1341	-0.3536	POS-biased
Enriched	relaxed, content, curious	-1.021000519059844	1.4156885263049306	0.1633588834462851	1.4583	2.3313	-0.873	POS-biased
Fighting	agitated, frustrated, tense	0.3392373830808953	2.676249869950163	0.0241196155260329	2.5368	2.8938	-0.357	POS-biased
Huddling	calm, relaxed, content	-0.2806031859098382	2.274656300208816	0.1158830058841413	2.0737	2.6443	-0.5706	POS-biased
LongIsolation	anxious, fearful, distressed	0.2284195402491188	2.053183944156107	0.1291938404496568	1.9035	2.3954	-0.4919	POS-biased
MissedNursing	distressed, uneasy, frustrated	-0.1102367738011859	2.734395879806116	0.1437010966322649	2.5347	3.0267	-0.492	POS-biased
NegativeConditioning	anxious, fearful, uneasy	-1.6159368631492783	0.0906674649019172	0.0456249190493753	1.4075	2.2755	-0.868	POS-biased
NovelObject	curious, interested, alert	0.2202691250787325	1.107815834771213	-0.0326940263968641	1.0146	1.558	-0.5434	POS-biased
PositiveConditioning	content, confident, relaxed	-1.6282264903645356	0.2769521783315541	0.1012287793643559	1.4159	2.3276	-0.9117	POS-biased
Restrain	anxious, fearful, agitated	0.7936762711520284	3.010519502863924	-0.1132149075059827	2.9922	3.1898	-0.1976	POS-biased
Run	active, playful, excited	-0.7956623561839257	1.9884008176716943	0.0666868324587822	1.879	2.5586	-0.6796	POS-biased
ShortIsolation	anxious, alert, uneasy	0.0700210849643234	1.866504140067802	0.1290665416555929	1.6887	2.2519	-0.5632	POS-biased
ShortReunion	content, relaxed, positive	0.336234633453303	3.098753801213133	0.2198099719276851	2.9506	3.3593	-0.4087	POS-biased
Surprise	alert, anxious, uneasy	-0.637847418600486	1.124501346229115	0.0596273178333053	1.0162	1.8912	-0.875	POS-biased

Supplementary Table S5.

Vocal-only OAAS metrics by context, including vOAAS coordinates, distances to POS/NEG vocal centroids, and bias values.

context	vOAAS1	vOAAS2	vOAAS3	d_pos	d_neg	delta_pos_minus_neg	bias
BeforeNursing	-0.1766108296283038	1.1376987066063906	-1.018266712758509	0.7605256309793993	2.7107304882376884	-1.950204857258289	POS-biased
PositiveConditioning	-3.518249581607717	-0.924885408300042	-0.7480438432212855	3.270749227212812	5.0978336998196925	-1.827084472606881	POS-biased
Surprise	-2.249873380400567	0.6768640229777358	0.5377717860617968	2.07282586527779	3.873194191869264	-1.8003683265914745	POS-biased
NegativeConditioning	-3.594737123917902	-1.069083495519792	-0.4966907433039824	3.4148351682181546	5.135548598628563	-1.720713430410408	POS-biased
Huddling	-0.2367513134854081	0.7683568220760217	-0.3583183004235747	0.5665033097122806	2.240244912285082	-1.6737416025728011	POS-biased
AfterNursing	-0.1204405794175916	2.1596554881388483	-0.7103187708675576	1.6506557743321315	3.289573630511858	-1.6389178561797266	POS-biased
Enriched	-1.812673404177451	-0.8133256980993299	-0.5869643728168832	1.846257807226326	3.404361129889174	-1.5581033226628478	POS-biased
Barren	-2.677553377080548	-1.267947191569744	-0.3151114697913092	2.8032567286547208	4.23791563475934	-1.4346589061046195	POS-biased
MissedNursing	-0.0324705398104772	1.019975773337449	-0.3482805806091255	0.8252362869997039	2.2527390250493355	-1.4275027380496317	POS-biased
Run	-0.6863372353411598	-1.351834154443547	-0.6369376605733024	1.9413638636790784	2.501674865713952	-0.5603110020348729	POS-biased
NovelObject	-1.1117103331088916	0.4897062607636988	2.648009148767731	3.421465986018279	3.5398535114052527	-0.1183875253869737	POS-biased
ShortIsolation	-0.2255433854245974	0.0414579743093919	1.339838113992697	2.1816467269239714	1.9848896895313848	0.1967570373925868	NEG-biased
ShortReunion	2.601699362999976	1.7390049646579784	-1.2772609588341228	3.4618761492072307	3.0475815350251825	0.4142946141820482	NEG-biased
LongIsolation	0.3738409134302822	0.3830388248812313	1.2679161371151957	2.244956463991956	1.637805930750184	0.6071505332417722	NEG-biased
Fighting	1.4016752552172371	0.2713125899703875	0.5933370525542233	2.437763261085337	0.8137404263845605	1.6240228347007764	NEG-biased
Castration	4.78990210321373	-1.2915818022220842	-0.6316302383131935	5.724770706689885	3.5977138535226216	2.127056853167263	NEG-biased
Restrain	5.092133729251422	-1.6096956087798238	0.2556981207260692	6.196877923413168	3.822390872050209	2.374487051362959	NEG-biased
Crushing	2.1836997192879664	-0.3587180687847711	0.4852532922951337	3.198350866239144	0.7735266822413499	2.424824183997793	NEG-biased

Supplementary Table S6.

Explained variance ratios for the vocal-only PCA configuration (vOAAS1–vOAAS3).

component	explained_variance_ratio
vOAAS1	0.6499071375342012
vOAAS2	0.1360944240839191
vOAAS3	0.1039714197937889

Supplementary Table S7.

Complete consolidated ranking across all stimuli (including noise references), reporting centroid-distance metrics and OAAS-derived indices.

file	domain	OAAS1	OAAS2	OAAS3	d_pos_euclid	d_neg_euclid	delta_euclid
W1_01.wav	music	-0.5954207642815079	-1.4797441140409884	0.4604206426898302	3.542626957731084	2.397114706503884	-1.1455122512272
W1_02.wav	music	-1.116817403868397	-1.4342238267193366	-0.3139037111596099	3.588282753453721	2.218357462519464	-1.369925290934258
W1_03.wav	music	-0.1694911472400826	-1.900905082404048	-0.2122287161532131	3.9408035206146814	2.209195347961757	-1.731608172652925
W1_04.wav	music	-0.442746035370773	-1.0783481122644452	-0.7821415887360239	3.2303814074046984	1.3982810960882668	-1.832100311316432
W1_05.wav	music	8.985567892323187	-0.8812156163891499	4.095816874619465	10.489552596477726	10.181443390624354	-0.3081092058533716
W1_06.wav	music	-1.4646175181297023	-2.071304201015194	0.5881270418921913	4.30081640037542	3.264137325480945	-1.036679074894475
W1_07.wav	music	-0.4527137649904876	-1.9973330426520173	0.6022435346229734	4.061948107845546	2.797008641313377	-1.2649394665321694
W1_08.wav	music	-0.2143447074545362	0.2020483861975167	0.1093471158262034	1.8260622836105915	1.3841388593663977	-0.4419234242441936
W2_01.wav	music	-0.8037534316209027	-0.1290502014635611	0.1416404997142027	2.224220379735707	1.7227617331761564	-0.5014586465595507
W2_02.wav	music	-1.320526282743118	-0.5611960931601331	0.6394307513890143	2.85070790967114	2.5023621090321404	-0.3483458006389996
W2_03.wav	music	-1.289235236184213	-0.4053410624885935	-0.0390592705250352	2.6439689710998526	2.004558494423339	-0.6394104766765136
W2_04.wav	music	-0.0225159753839707	-0.6016116350690116	-0.762103047825078	2.773296456663986	0.8226819629355578	-1.950614493728428
W2_05.wav	music	0.0135347625511813	0.0928550803814571	-0.0895012506068431	1.962004915391749	1.1125536919882686	-0.8494512234034803
W2_06.wav	music	-0.3127923944003612	-1.5852189185404453	0.52690774265012	3.6395348631431457	2.4213348806400967	-1.218199982503049
W2_07.wav	music	-0.7085055580404035	-0.6062736230167382	0.5913361004655153	2.7183490244816406	2.126356942354424	-0.5919920821272173
W3_01.wav	music	-0.8033387393291499	-0.6403811691533136	-0.1593848456902387	2.7334451708137446	1.6444601040779132	-1.0889850667358314
W3_02.wav	music	-1.3025365816949095	-0.589676287494638	0.6434697846277757	2.870798366023101	2.500540541150577	-0.3702578248725237
W3_03.wav	music	-0.4634530331702701	-1.273324040754759	1.1321401786169232	3.4678340179200107	2.750982108569485	-0.7168519093505257
W3_04.wav	music	-0.9843388415244684	-0.8774035583197483	-0.4584261739162106	3.0428250167207653	1.734744788413452	-1.3080802283073132
W3_05.wav	music	-0.1635679048250887	0.2168373706419207	-1.5717503166116769	2.458061684679816	0.6561427455493342	-1.8019189391304813
W3_06.wav	music	-0.6196028895434451	-1.8924368275484795	0.4751291112525701	3.95514340238341	2.6943777960123594	-1.2607656063710513
W3_07.wav	music	-0.968108949569936	-2.026231713886353	-0.5018863344362862	4.1567958254507245	2.515182680022906	-1.6416131454278189
W4_01.wav	music	-0.8037534316209027	-0.1290502014635611	0.1416404997142027	2.224220379735707	1.7227617331761564	-0.5014586465595507
W4_02.wav	music	-1.320526282743118	-0.5611960931601331	0.6394307513890143	2.85070790967114	2.5023621090321404	-0.3483458006389996
W4_03.wav	music	-1.289235236184213	-0.4053410624885935	-0.0390592705250352	2.6439689710998526	2.004558494423339	-0.6394104766765136
W4_04.wav	music	-0.0225159753839707	-0.6016116350690116	-0.762103047825078	2.773296456663986	0.8226819629355578	-1.950614493728428
W4_05.wav	music	0.0135347625511813	0.0928550803814571	-0.0895012506068431	1.962004915391749	1.1125536919882686	-0.8494512234034803
W4_06.wav	music	-0.3127923944003612	-1.5852189185404453	0.52690774265012	3.6395348631431457	2.4213348806400967	-1.218199982503049
W4_07.wav	music	-0.7085055580404035	-0.6062736230167382	0.5913361004655153	2.7183490244816406	2.126356942354424	-0.5919920821272173

file	domain	OAAS1	OAAS2	OAAS3	d_pos_euclid	d_neg_euclid	delta_euclid
W5_01.wav	music	-0.8033387393291499	-0.6403811691533136	-0.1593848456902387	2.7334451708137446	1.6444601040779132	-1.0889850667358314
W5_02.wav	music	-1.3025365816949095	-0.589676287494638	0.6434697846277757	2.870798366023101	2.500540541150577	-0.3702578248725237
W5_03.wav	music	-0.4634530331702701	-1.273324040754759	1.1321401786169232	3.4678340179200107	2.750982108569485	-0.7168519093505257
W5_04.wav	music	-0.9843388415244684	-0.8774035583197483	-0.4584261739162106	3.0428250167207653	1.734744788413452	-1.3080802283073132
W5_05.wav	music	-0.1635679048250887	0.2168373706419207	-1.5717503166116769	2.458061684679816	0.6561427455493342	-1.8019189391304813
W5_06.wav	music	-0.6196028895434451	-1.8924368275484795	0.4751291112525701	3.95514340238341	2.6943777960123594	-1.2607656063710513
W5_07.wav	music	-0.968108949569936	-2.026231713886353	-0.5018863344362862	4.1567958254507245	2.515182680022906	-1.6416131454278189
W6_01.wav	music	7.277017284417674	-0.6294403762311236	-1.8754832291987749	8.230557263964494	7.03176411911929	-1.1987931448452036
W6_02.wav	music	-1.386751458638167	-0.1064284185160125	-0.1440604965374754	2.4234742234449103	1.9854036738413328	-0.438070549603577
W6_03.wav	music	7.617793442096421	-1.1090519717088372	-1.920560092709888	8.715136774970958	7.432525171889743	-1.2826116030812145
W6_04.wav	music	-1.482433907925715	-0.9538487605783312	-0.4433888237194514	3.265083058300733	2.17188560405689	-1.093197454243843
W6_05.wav	music	0.4675865255883913	-1.2495465712761242	-0.5988446839180059	3.426564039525259	1.4105395751205871	-2.016024464404672
W6_06.wav	music	-0.3885974671064289	-1.479600900163366	1.0212453999887807	3.63128030516657	2.7466043981149064	-0.8846759070516637
W6_07.wav	music	-0.899469337447385	-0.9705398166670112	0.6696546169451779	3.119777368272758	2.414312323809771	-0.7054650444629873
W7_01.wav	music	-0.7274312719607853	-0.5314779488856917	0.8126145036749107	2.70001636302573	2.300811728657212	-0.3992046343685183
W7_02.wav	music	1.5825960757605202	-0.7554031444199829	-0.0042033345381354	3.339590280668501	1.8896198902788648	-1.449970390389636

Supplementary Table S7 (continued).

file	domain	OAAS1	OAAS2	OAAS3	d_pos_euclid	d_neg_euclid	delta_euclid
W7_03.wav	music	-1.1417109116876345	-1.9316012238768925	-0.6432030174694235	4.12102792081325	2.500827969760264	-1.620199951052987
W7_04.wav	music	-0.7825117616195572	-1.9332218597752429	0.322457762501717	4.001607620315145	2.6969266246424413	-1.3046809956727037
W7_05.wav	music	0.5256172629910854	-1.1437019224060527	0.8200735166472749	3.3483281068389177	2.31384714786047	-1.0344809589784476
W7_06.wav	music	-0.5072876450607992	-1.1108556925972042	0.8298725243644237	3.234109615325316	2.438256514420777	-0.7958531009045382
W7_07.wav	music	1.915285260665352	-2.674754383630324	-1.5536603353499985	5.435341786395151	3.1729349007755183	-2.2624068856196327
W8_01.wav	music	-0.899469337447385	-0.9705398166670112	0.6696546169451779	3.119777368272758	2.414312323809771	-0.7054650444629873
W8_02.wav	music	7.277017284417674	-0.6294403762311236	-1.8754832291987749	8.230557263964494	7.03176411911929	-1.1987931448452036
W8_03.wav	music	-1.386751458638167	-0.1064284185160125	-0.1440604965374754	2.4234742234449103	1.9854036738413328	-0.438070549603577
W8_04.wav	music	7.617793442096421	-1.1090519717088372	-1.920560092709888	8.715136774970958	7.432525171889743	-1.2826116030812145
W8_05.wav	music	-1.482433907925715	-0.9538487605783312	-0.4433888237194514	3.265083058300733	2.17188560405689	-1.093197454243843
W8_06.wav	music	0.4675865255883913	-1.2495465712761242	-0.5988446839180059	3.426564039525259	1.4105395751205871	-2.016024464404672
W8_08.wav	music	-0.3885974671064289	-1.479600900163366	1.0212453999887807	3.63128030516657	2.7466043981149064	-0.8846759070516637
W9_01.wav	music	-0.8033387393291499	-0.6403811691533136	-0.1593848456902387	2.7334451708137446	1.6444601040779132	-1.0889850667358314
W9_02.wav	music	-1.3025365816949095	-0.589676287494638	0.6434697846277757	2.870798366023101	2.500540541150577	-0.3702578248725237
W9_03.wav	music	-0.4634530331702701	-1.273324040754759	1.1321401786169232	3.4678340179200107	2.750982108569485	-0.7168519093505257
W9_04.wav	music	-0.9843388415244684	-0.8774035583197483	-0.4584261739162106	3.0428250167207653	1.734744788413452	-1.3080802283073132
W9_05.wav	music	-0.1635679048250887	0.2168373706419207	-1.5717503166116769	2.458061684679816	0.6561427455493342	-1.8019189391304813
W9_06.wav	music	-0.6196028895434451	-1.8924368275484795	0.4751291112525701	3.95514340238341	2.6943777960123594	-1.2607656063710513
W9_07.wav	music	-0.968108949569936	-2.026231713886353	-0.5018863344362862	4.1567958254507245	2.515182680022906	-1.6416131454278189
W10_01.wav	music	-0.5954207642815079	-1.4797441140409884	0.4604206426898302	3.542626957731084	2.397114706503884	-1.1455122512272
W10_02.wav	music	-1.116817403868397	-1.4342238267193366	-0.3139037111596099	3.588282753453721	2.218357462519464	-1.369925290934258
W10_03.wav	music	-0.1694911472400826	-1.900905082404048	-0.2122287161532131	3.9408035206146814	2.209195347961757	-1.731608172652925
W10_04.wav	music	-0.442746035370773	-1.0783481122644452	-0.7821415887360239	3.2303814074046984	1.3982810960882668	-1.832100311316432
W10_05.wav	music	8.985567892323187	-0.8812156163891499	4.095816874619465	10.489552596477726	10.181443390624354	-0.3081092058533716
W10_06.wav	music	-1.4646175181297023	-2.071304201015194	0.5881270418921913	4.30081640037542	3.264137325480945	-1.036679074894475
W10_07.wav	music	-0.4527137649904876	-1.9973330426520173	0.6022435346229734	4.061948107845546	2.797008641313377	-1.2649394665321694
W10_08.wav	music	-0.2143447074545362	0.2020483861975167	0.1093471158262034	1.8260622836105915	1.3841388593663977	-0.4419234242441936
Negaive_01.wav	challenge_reference	1.5959684207530938	0.9851981053842336	-0.5803589404056136	2.2322000391521275	1.6975092078536045	-0.534690831298523
Negative_02.wav	challenge_reference	0.1399034583341609	-0.1177725659323337	1.067329516639196	2.3920784301860403	2.238674530238355	-0.1534038999476852
Negative_02_part1.wav	challenge_reference	-0.6094601672414707	0.4556537332638108	-0.5437084360421973	1.7292627378008452	1.1862756400075136	-0.5429870977933315

file	domain	OAAS1	OAAS2	OAAS3	d_pos_euclid	d_neg_euclid	delta_euclid
Negative_02_part2.wav	challenge_reference	-1.1199693162031192	0.0046061675004494	-0.3508074236909154	2.2407844061521227	1.6460614778220215	-0.594722928330101
Negative_02_part3.wav	challenge_reference	-0.0030316027199865	-0.5184714444546679	-3.056790787262164	4.054280772841859	2.0019077431784424	-2.052373029663417
Negative_03.wav	challenge_reference	1.877633713339855	-0.5952803237737669	-3.4916068900313624	4.926142143957697	2.878213944578974	-2.047928199378723
AfterNursing__SW__01.wav	vocalization	-0.082980563405053	2.855170533451856	0.276666356461388	0.867599028287396	3.188876748620112	2.3212777203327155
AfterNursing__SW__02.wav	vocalization	-0.0831842857693972	2.841607223418773	0.2734965236174368	0.8539305812188668	3.175484673213184	2.321554091994317
Barren__SW__01.wav	vocalization	-1.3065006961442365	0.6643889184480695	0.0291689587021219	1.7185199367749138	2.105290231959115	0.3867702951842013
Barren__SW__02.wav	vocalization	-1.300882927710072	0.6542084321209958	0.027344342639029	1.7232741236522855	2.096913236885613	0.3736391132333274
BeforeNursing__SW__01.wav	vocalization	-0.2922050822597513	3.0221539943609432	0.1409542140994785	0.996665498578576	3.31312287790828	2.3164573793297043
BeforeNursing__SW__02.wav	vocalization	-0.2954382376960834	3.012921388784592	0.1374162268421566	0.9873727709769248	3.3040068225686614	2.3166340515917367
Castration__SW__01.wav	vocalization	0.6501430349124923	3.3560331761790763	-0.0024176783316264	1.6138435745556354	3.532231949327171	1.918388374771536
Castration__SW__02.wav	vocalization	0.646460221696735	3.327909944181866	-0.0050195230100986	1.588812643647748	3.5045990616918368	1.915786418044089
Crushing__SW__01.wav	vocalization	0.4224627460523301	2.9072420489595765	0.0977682149861981	1.1144506976697235	3.1365841818824696	2.0221334842127456
Crushing__SW__02.wav	vocalization	0.4331411998241004	2.907354065862044	0.0841649333502178	1.121086236841173	3.131649493216837	2.010563256375664
Enriched__SW__01.wav	vocalization	-1.0227072966314152	1.4043788853636814	0.1616851798272741	0.9864480335215308	2.324637166842161	1.3381891333206302
Enriched__SW__02.wav	vocalization	-1.0192937414882723	1.4269981672461796	0.1650325870652962	0.969911077669244	2.337971973088884	1.3680608954196392

Supplementary Table S7 (continued).

file	domain	OAAS1	OAAS2	OAAS3	d_pos_euclid	d_neg_euclid	delta_euclid
Fighting__SW__01.wav	vocalization	0.3454602151112547	2.67533596950557	0.0211066991673867	0.8903118806786577	2.891772743094827	2.0014608624161694
Fighting__SW__02.wav	vocalization	0.333014551050536	2.677163770394756	0.0271325318846792	0.8827834716848285	2.8957942900331464	2.013010818348318
Huddling__SW__01.wav	vocalization	-0.278840852667629	2.266418880509072	0.1134171055007872	0.2410282923381518	2.6357258420232563	2.3946975496851044
Huddling__SW__02.wav	vocalization	-0.2823655191520475	2.2828937199085586	0.1183489062674955	0.2581732865240815	2.652845883829655	2.3946725973055734
LongIsolation__SW__01.wav	vocalization	0.1692976943986189	1.9745326409514723	0.1236954113621963	0.4354536053154258	2.32941600904446	1.893962403729034
LongIsolation__SW__02.wav	vocalization	0.2875413860996189	2.131835247360742	0.1346922695371173	0.5607991734654783	2.463558596032138	1.9027594225666595
MissedNursing__SW__01.wav	vocalization	-0.113320859891498	2.740464094052476	0.1468041776667622	0.7306889202822652	3.033832834419855	2.3031439141375896
MissedNursing__SW__02.wav	vocalization	-0.1071526877108738	2.728327665559757	0.1405980155977676	0.7196650141086373	3.0194758665479657	2.2998108524393284
NegativeConditioning__SW__01.wav	vocalization	-1.6115699490083664	0.0792350099618692	0.042233567451418	2.370687413930449	2.2697053120300263	-0.1009821019004229
NegativeConditioning__SW__02.wav	vocalization	-1.62030377729019	0.1020999198419652	0.0490162706473327	2.3568367899581006	2.281255438360816	-0.0755813515972847
NovelObject__SW__01.wav	vocalization	0.216452309588236	1.0989827816242983	-0.0350640291613968	1.0513935871103584	1.5504981316538151	0.4991045445434567
NovelObject__SW__02.wav	vocalization	0.224085940569229	1.116648887918128	-0.0303240236323314	1.0388213064900231	1.5656255039787652	0.5268041974887421
PositiveConditioning__SW__01.wav	vocalization	-1.6288973252546903	0.2733971687476041	0.1003686077749693	2.224074050538995	2.3272907560125806	0.1032167054735855
PositiveConditioning__SW__02.wav	vocalization	-1.6275556554743809	0.2805071879155043	0.1020889509537426	2.2176547576939303	2.3278402746297764	0.1101855169358461
Restrain__SW__01.wav	vocalization	0.7580423219050108	3.035057111615391	-0.1028619439210101	1.4460908454770784	3.210939480421329	1.7648486349442507
Restrain__SW__02.wav	vocalization	0.829310220399046	2.985981894112458	-0.1235678710909554	1.4674484523102906	3.169111563511356	1.7016631112010647
Run__SW__01.wav	vocalization	-0.7965101039324537	1.9810930271926936	0.0636948548834376	0.5376006486853676	2.551987525545069	2.0143868768597013
Run__SW__02.wav	vocalization	-0.7948146084353976	1.9957086081506952	0.0696788100341269	0.5346118103780875	2.5652744380883896	2.030662627710302
ShortIsolation__SW__01.wav	vocalization	0.0828167734616437	1.846764320874256	0.1244169251414676	0.3904817366832583	2.23188782130818	1.8414060846249207
ShortIsolation__SW__02.wav	vocalization	0.0572253964670032	1.886243959261348	0.1337161581697182	0.3515302579074235	2.2720717656287106	1.920541507721287
ShortReunion__SW__01.wav	vocalization	0.3374940284454479	3.1051500427349463	0.2221975053199616	1.2403098077658483	3.3661510080385995	2.1258412002727507
ShortReunion__SW__02.wav	vocalization	0.3349752384611582	3.092357559691321	0.2174224385354086	1.22747163022476	3.352509201101768	2.125037570877008
Surprise__SW__01.wav	vocalization	-0.6391467493080365	1.117535111020706	0.0579405611954019	0.9855771807860176	1.886728863038448	0.9011516822524306
Surprise__SW__02.wav	vocalization	-0.6365480878929355	1.131467581437524	0.0613140744712089	0.9716258627579484	1.895617636051964	0.9239917732940154

Supplementary Table S8. Microdynamic metrics for OAAS-directed transformations (W7_01 and W8_03).

Peak level (dBFS), RMS amplitude, RMS drift relative to the original stimulus, crest factor (peak-to-RMS ratio, dB), high-percentile amplitude statistics (p95, p99), and dynamic range proxy (p95–p10, dB) are reported for original, POS-directed, and NEG-directed variants. Transformations were RMS-preserving (v6 pipeline), enabling isolation of microdynamic reorganization independent of global energy changes.

stim	variant	peak_dBFS	RMS	RMS_drift_%	crest_dB	p95	p99	DR_p95_p10_dB
W7_01	Original	-7.098	0.086122	0.00	14.20	0.179626	0.257172	26.96
W7_01	POS	-11.401	0.085942	-0.21	9.92	0.173309	0.224030	22.38
W7_01	NEG	-1.000	0.082721	-3.95	20.65	0.175049	0.288910	29.88
W8_03	Original	-10.533	0.055455	0.00	14.59	0.109589	0.147064	24.58
W8_03	POS	-16.990	0.055632	+0.32	8.10	0.088257	0.101898	17.93
W8_03	NEG	-7.901	0.055364	-0.16	17.23	0.114136	0.164734	26.50

Interpretive summary: POS-directed variants exhibit reduced crest factor and narrower high-percentile amplitude dispersion relative to the original stimuli, whereas NEG-directed variants show increased crest factor and expanded percentile dispersion, consistent with directional microdynamic reorganization under RMS-preserving constraints.